

# Allampati Bharath Kumar Reddy

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## ABOUT ME

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Computer Science graduate with hands-on experience building and deploying full-stack applications using Java, JavaScript, React, Node.js, SQL, and cloud platforms. Strong foundation in object-oriented programming, data structures, databases, and software development fundamentals. Built production-style projects featuring authentication, REST-based integrations, AI capabilities, real-time features, and cloud deployment. Seeking a Software Engineering role focused on building scalable, reliable, and high-quality applications.

## PROJECTS

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### SeatSync — High-Concurrency Ticket Booking Microservices — [link](#) ( Present )

*Java 17+, Spring Boot 3.3.2, Spring Cloud, PostgreSQL, Redis, Apache Kafka, Docker, React.js, TypeScript, Tailwind CSS v4*

- Developed a high-concurrency ticket booking platform using a Spring Boot microservices architecture (Eureka Registry, API Gateway, Event/Booking/User services).
- Implemented Redis-based distributed holds (TTL-based seat reservation for 5 minutes) and database-level JPA Optimistic Locking to guarantee zero double bookings under heavy traffic.
- Designed an idempotent payment processing workflow using unique transaction keys to prevent duplicate charges.
- Built a responsive React/TypeScript frontend featuring a real-time countdown timer, dynamic interactive seating grid, and JWT-based authentication with silent auto-refresh loops.

### Aether Cast — Podcast Streaming Platform — [link](#)

*React.js, Node.js, Express.js, MongoDB Atlas, Supabase Auth, Gemini API*

- Built a full-stack podcast platform with React 18, Node.js/Express, MongoDB Atlas, and Supabase authentication, featuring a global sticky audio player, PWA offline support, and live synced transcripts. Added automated API tests for authentication, validation, and core backend workflows.
- Integrated Google Gemini 1.5 Flash API for AI-powered episode summaries and smart tagging, along with real-time listening parties, creator drafts workspace, and in-browser recording with waveform visualization.
- Deployed frontend on Vercel and backend on Render with a full admin command center for platform-wide metrics, user management, and content moderation.

### GATE Flow — [link](#)

*React.js, JavaScript, CSS, Firebase, Gemini API, AI Integration*

- Developed a comprehensive GATE CS & IT preparation platform featuring topic-wise practice, customizable CBT-style mock tests, performance analytics, and an AI-powered study assistant.
- Built configurable mock test workflows with subject selection, question navigation, negative marking, mistake analysis, and an integrated calculator aligned with GATE exam patterns.
- Designed modular workflows for topic-wise practice, configurable mock tests, question navigation, scoring, negative marking, and performance analysis.

### Digital Watermarking System — [link](#)

*Python, Image Processing, LSB, DCT, QR Encoding, ML*

- Implemented a robust digital watermarking framework supporting both LSB-based and DCT-based embedding techniques for securely hiding QR-encoded data inside color images.
- Designed end-to-end watermark workflows including QR generation, bit-level encoding, block-wise DCT coefficient manipulation, and image reconstruction with minimal perceptual distortion.
- Evaluated watermark quality using PSNR and tested extraction robustness under compression, noise, and format transformations.

## SKILLS

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- **Programming Languages:** Java , JavaScript, TypeScript, SQL, Python(basics)
- **Core Java:** OOP, Collections Framework, Exception Handling, Stream API
- **Frontend Technologies:** HTML5, CSS3, JavaScript, TypeScript, React.js, Tailwind CSS, Bootstrap
- **Backend & Framework:** Spring Boot, Spring Cloud, Spring Framework, Microservices, Node.js, Express.js
- **Database:** MySQL, PostgreSQL, MongoDB, Redis
- **Version Control & Tools:** Git, GitHub, Docker, Docker Compose, Apache Kafka
- **Currently Learning:** RESTful APIs, Multithreading
- **Relevant Coursework:** Data Structures & Algorithms, Database Management, Operating Systems

EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Vellore Institute of Technology</b><br>B. Tech in Computer Science (Specialization: Information Security),<br>8.39/10 CGPA | 2022 – 2026<br>Vellore, Tamil Nadu     |
| <b>Narayana Junior College</b><br>Class 12(State Board), 92.2% Higher Secondary                                               | 2020 – 2022<br>Nellore, Andhra Pradesh |
| <b>Narayana High School</b><br>Class 10(State Board), 99.5 %                                                                  | 2019 – 2020<br>Nellore, Andhra Pradesh |

CERTIFICATES AND TRAINING

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| Oracle OCI Generative AI Professional -- Oracle   2025   <a href="#">Credential</a> | May 2025 – June 2025 |
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